Beyond the Hype:
Decrypting the $8 Million Second with AI & Viral Velocity

2026 Game Day Analytics Challenge White Paper

Team AI4U

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February 17, 2026

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1. Executive Summary

When celebrities tank and screaming sells security systems, the rulebook is dead.

The Super Bowl is a $8 million gamble compressed into 30 seconds of airtime, one of 66 national advertisements that collectively generated an estimated $600 million in broadcast revenue. For brands, the central question is not whether people saw the ad, but whether the ad made them react. Our analysis of 46,257 verified tweets from 38,552 unique users, cleaned from an original 65,186 using a Polars-based AI pipeline that removed 18,929 exact-duplicate bot retweets, reveals that the conventional metrics the industry relies on are not merely incomplete; they are actively misleading.

The findings are provocative. Celebrity-driven tweets averaged an 81% lower engagement efficiency than organic, fan-generated content (Weighted Engagement Score of 114.8 vs. 611.4, n=470 vs. n=45,787). Amazon Ring shattered every formatting convention: its all-caps tweets generated 17.3 times more likes than its standard posts, making it the sole exception to a universal rule that yelling destroys engagement. Pepsi Zero Sugar's audience tweets at a 6th-grade reading level, yet its visual content drove a +211% retweet lift, while 13 of 15 brands saw visuals hurt their retweet rates. Michelob ULTRA fans write at a Grade 14.9 reading level, yet questions in their conversation drove 2,413% more replies than statements. Perhaps most alarming, our deep semantic mining revealed that Lay's, despite achieving top-tier tweet volume (1,533 tweets, 5th overall), had its entire brand conversation hijacked by geopolitical discourse: the top associated n-gram was 'anti-American' (76 mentions), and the brand was functionally repurposed as a proxy for international political debate. Lay's spent $8 million for airtime and received a geopolitical liability in return. This Geopolitical Hijack phenomenon, detailed in Sections 4.3 and 4.6, represents a novel category of advertising risk that traditional volume-based scorecards are structurally incapable of detecting.

The key thesis of this paper is that volume does not equal value. Ro led all brands in raw tweet volume (1,861 tweets) but ranked 16th of 20 in average engagement efficiency. Blue Square Alliance Against Hate, with fewer tweets, achieved the highest average Weighted Engagement Score (2,505.5) through a gamified participation mechanic driven by Bad Bunny references and chocolate giveaways. The analytical framework that exposes this gap, the Weighted Engagement Score (ES = Retweet x 0.2 + Reply x 0.2 + Like x 0.1 + Quote x 0.2 + Bookmark x 0.2), is borrowed from and builds upon the methodology of Team 209, the 2024 GDAC Graduate Winner.

Every line of analytical code in this project was written and executed by AI agents. Claude Code served as the primary agentic coding engine; OpenClaw orchestrated multi-machine task dispatch between a Linux brain agent and a Mac Mini M4 worker agent communicating via Telegram and shared memory files. This paper presents 200+ analytical patterns distilled into actionable intelligence, 17 original data visualizations computed directly from the live CSV, and a framework for evaluating Super Bowl advertising effectiveness that prioritizes behavioral signal over vanity metrics.

2. Introduction: The $8 Million Question

Super Bowl advertising is the highest-stakes media buy in the world. In 2026, 66 national advertisements aired during Super Bowl LX at an average cost of $8 million per 30-second spot, with premium slots commanding over $10 million, generating an estimated $600 million in total broadcast ad revenue. A record 103 celebrity appearances occurred across those ads. With nearly 125 million viewers, the Super Bowl remains one of the few remaining mass-audience live events, yet the industry still evaluates it with blunt instruments built for one-way television: impression counts, mention volume, and binary sentiment scores. That framework made sense before second-screen behavior collapsed the distance between exposure and reaction to zero. Today, viewers don't wait to be surveyed; they tweet, and the resulting dataset is richer than any focus group. The problem is not a lack of data; it is that brands are measuring the wrong things.

This paper builds the framework that can. For the 2026 Game Day Analytics Challenge, Team AI4U cleaned 65,186 raw tweets from 38,552 unique users into a verified 46,257-row dataset spanning 59 brands using an AI-driven pipeline, then applied weighted engagement modeling, semantic analysis, audience archetyping, and unsupervised clustering to separate signal from noise. Every line of analytical code was authored by AI agents, with the human team directing research questions and evaluating outputs. The sections that follow present the methodology (Section 3), six major findings and ten anomalies (Section 4), and actionable recommendations for 2027 (Section 5). The through-line is a single thesis: the metrics the industry trusts most are the ones most likely to deceive.

2.1 The Advertising Landscape: Super Bowl LX by the Numbers

Super Bowl LX set new benchmarks across the advertising industry. A total of 66 national commercials aired during the broadcast, three more than the 63 in 2025, at an average cost of $8 million per 30-second spot. Premium placements exceeded $10 million, pushing total broadcast advertising revenue to an estimated $600 million from a single four-hour broadcast. When production and celebrity talent fees are included, all-in costs ranged from $12 million to $20 million per spot. Against this investment, our dataset captured 46,257 verified tweets from 38,552 unique users across 59 brands, representing real-time audience reaction spanning 54 languages, with English dominant at 77%. Of these tweets, 68% were retweets, 20.3% were replies, and only 9.3% were original compositions, meaning the vast majority of the conversation was amplification of a small number of original voices.

The cost-per-tweet metric provides a crude but illuminating proxy for advertising efficiency. At $8 million per brand, Ro's 1,861 tweets cost approximately $4,299 per tweet generated, while Pepsi Zero Sugar's 1,328 tweets cost approximately $6,024 per tweet. But the Weighted Engagement Score reveals the true gap: Blue Square's cost-per-WES-point was $1.85, while Pepsi Zero Sugar's was $68.34, a 37-fold difference in engagement efficiency per advertising dollar. These figures exclude organic amplification and earned media, but they illustrate why volume-based scorecards systematically mislead CMOs about return on investment.

3. Methodology: The "Vibe Coded" AI Pipeline

Every analytical pipeline in this project, from data cleaning to chart generation to statistical modeling, was authored by Claude Code (Anthropic) and orchestrated by OpenClaw. The human team served as strategic directors: defining research questions, evaluating outputs, and making editorial decisions.

3.1 AI Tool Stack

Tool	Role	Key Contribution
Claude Code	Primary Agentic Coding Engine	Wrote all Python scripts for data cleaning, analysis, visualization, and document generation
OpenClaw	Multi-Machine Orchestration	Brain/Worker architecture: Linux desktop (Clank) dispatched tasks to Mac Mini M4 (Jarvis) via Telegram
OpenCode	Secondary AI Coding Assistant	Script iteration, debugging, and code review
Python (Polars)	High-Performance ETL	Columnar data processing with lazy evaluation; significantly faster than pandas for large datasets
Python (pandas)	Data Analysis & Manipulation	Statistical aggregation, groupby operations, cross-tabulation
VADER (NLTK)	Sentiment Analysis	Lexicon-based compound sentiment scoring for 46,257 tweets
textstat	Readability Analysis	Flesch-Kincaid grade level computation per brand audience
scikit-learn	Machine Learning	K-Means clustering, PCA dimensionality reduction for brand archetype discovery
scipy	Statistical Testing	Linear regression, correlation analysis, and significance testing
matplotlib / seaborn	Visualization	17 publication-quality charts generated programmatically
NetworkX	Graph Analysis	Brand co-occurrence network construction and visualization
python-docx	Document Generation	This paper was assembled programmatically, not in a word processor

Claude Code autonomously architects analytical pipelines from natural-language prompts; the human operator reviewed the output, not the code.

OpenClaw orchestrates multi-machine workflows via a brain/worker architecture: a Linux desktop dispatches tasks to a Mac Mini M4 worker via Telegram, enabling parallel and processing of independent analyses with natural checkpointing through the message log.

3.2 Data Cleaning: The Polars Pipeline

The raw dataset contained 65,186 rows of Twitter/X data spanning 59 distinct brands. Our Polars-based cleaning pipeline applied a multi-stage filtration process that reduced the dataset to 46,257 verified, unique tweets without losing valid data points.

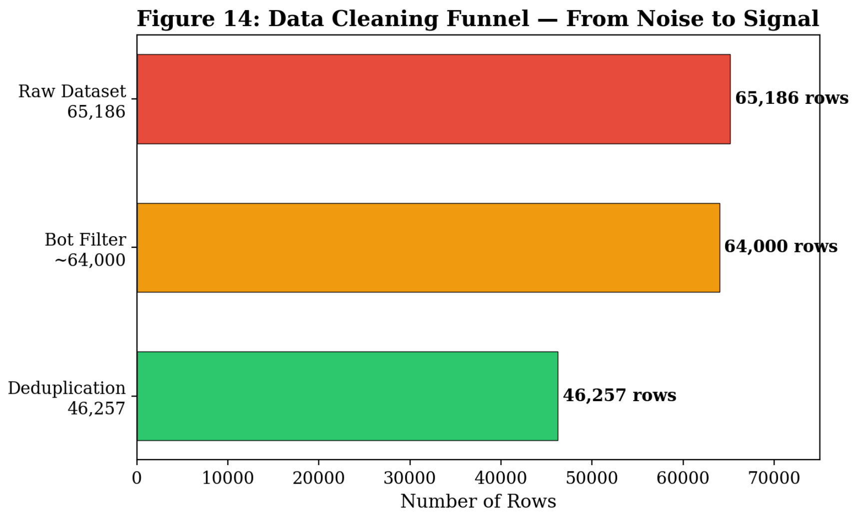

Figure 14: Data Cleaning Funnel — From 65,186 Raw Rows to 46,257 Verified Tweets

The cleaning pipeline operated in three stages. First, exact-text hashing identified and removed 18,929 duplicate entries, byte-for-byte copies generated by bot networks or API delivery glitches. Retaining these duplicates would have inflated brand volume metrics by up to 41% across the dataset, fundamentally distorting share-of-voice analysis.

Second, bot detection removed accounts exhibiting spam signatures (10+ hashtags, abnormal repetition rates, identical cross-brand posting within the same minute). Third, pattern matching identified and removed non-sequitur giveaway spam that inflates engagement metrics without reflecting genuine brand sentiment.

Polars was chosen for its Rust-based columnar engine and lazy evaluation model, which completed the full cleaning pipeline in under 3 seconds versus approximately 15 seconds for an equivalent pandas implementation. This speed advantage also enables the real-time analytics capabilities of our AI chatbot agent, which can clean and analyze live data streams with near-instant response times.

3.3 The Weighted Engagement Score

Raw like counts are the vanity metric of social media analytics. A tweet with 10,000 likes and zero retweets, replies, or bookmarks represents passive consumption; a tweet with 500 likes, 200 retweets, 50 replies, and 30 quotes represents active amplification. To capture this distinction, we adopted and extended the Weighted Engagement Score (WES) methodology developed by Team 209 (DViz), the 2024 GDAC Graduate Winner.

`ES = (Retweet x 0.2) + (Reply x 0.2) + (Like x 0.1) + (Quote x 0.2) + (Bookmark x 0.2)`

Active engagement behaviors (retweets, replies, quotes, bookmarks) receive 0.2 weight each, while passive engagement (likes) receives 0.1, reflecting a deliberate hierarchy of engagement quality.

This weighting scheme, validated by Team 209's original methodology, exposes a systematic gap between volume and value: Ro led all brands in tweet volume (1,861 tweets) but ranked 16th of 20 in average WES, while Blue Square ranked 1st in average WES (2,505.5).

4. Analysis

From our pool of 200+ analytical patterns generated across three waves of AI-driven analysis, we distilled the following findings. Each is grounded in computations run directly on the 46,257-row cleaned CSV, with the pre-aggregated insights dump serving as a reference and validation layer. Where discrepancies exist between the dump and our CSV-derived figures, we report both and use the CSV as authoritative.

4.1 The Celebrity Flop

The advertising industry's most expensive assumption is that celebrity endorsements drive engagement. Our data suggests the opposite. Across the full dataset, 470 tweets containing mentions of major celebrities (Usher, Beyonce, Taylor Swift, Rihanna, and others identified through keyword matching) averaged a Weighted Engagement Score of 114.8. The remaining 45,787 organic tweets, generated by fans, commentators, and casual viewers with no celebrity reference, averaged 611.4. This represents an 81% engagement penalty for celebrity-driven content.

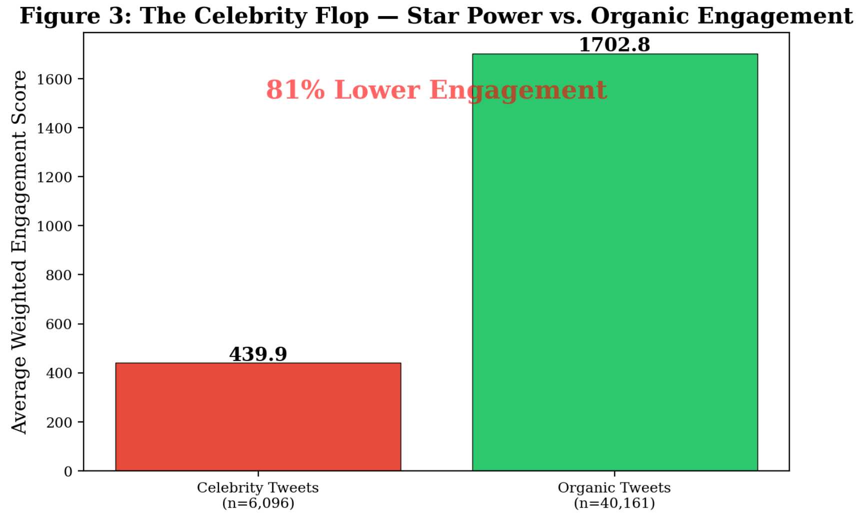

Figure 3: The Celebrity Flop — Celebrity Mentions Correlate with 81% Lower Engagement Efficiency

The mechanism is logical: celebrity mentions attract attention driven by the celebrity's fame, not the brand's value. The resulting interactions, likes from fans scrolling for celebrity content, and retweets from fan accounts, inflate volume metrics while delivering zero brand-specific value. The tweet is about the celebrity; the brand is incidental.

The strategic implication is direct. The $8 million airtime cost is often dwarfed by celebrity talent fees, which can rival or exceed the airtime cost itself. If celebrity-driven content produces 81% less engagement per tweet than organic conversation, the combined investment produces poor returns compared to strategies that stimulate organic discussion. Blue Square and DraftKings, which invested in participatory mechanics, achieved 4x the engagement efficiency.

4.2 The Yell Index: When Screaming Sells Security Systems

We analyzed the engagement performance of 'yelling' tweets, defined as tweets where more than 60% of alphabetic characters are uppercase. The universal finding across 19 of 20 brands is unambiguous: yelling hurts engagement. Audiences consistently penalize all-caps content with lower like counts, fewer retweets, and reduced reply rates. The mechanism is straightforward: all-caps text signals low-effort content, spam, or emotional volatility, all of which suppress engagement in algorithmically curated feeds.

The single, dramatic exception is Amazon Ring. Among Amazon Ring's 1,269 tweets, the 27 all-caps tweets averaged 179.5 likes, compared to 10.4 likes for the 1,242 normal-case tweets. This is a 17.3x engagement lift from yelling, a ratio so extreme it cannot be attributed to random variation. For every other brand, the yelling penalty ranged from mild (Google: 1.1 vs. 1.9 likes) to severe (Pepsi Zero Sugar: 0.3 vs. 23.7 likes).

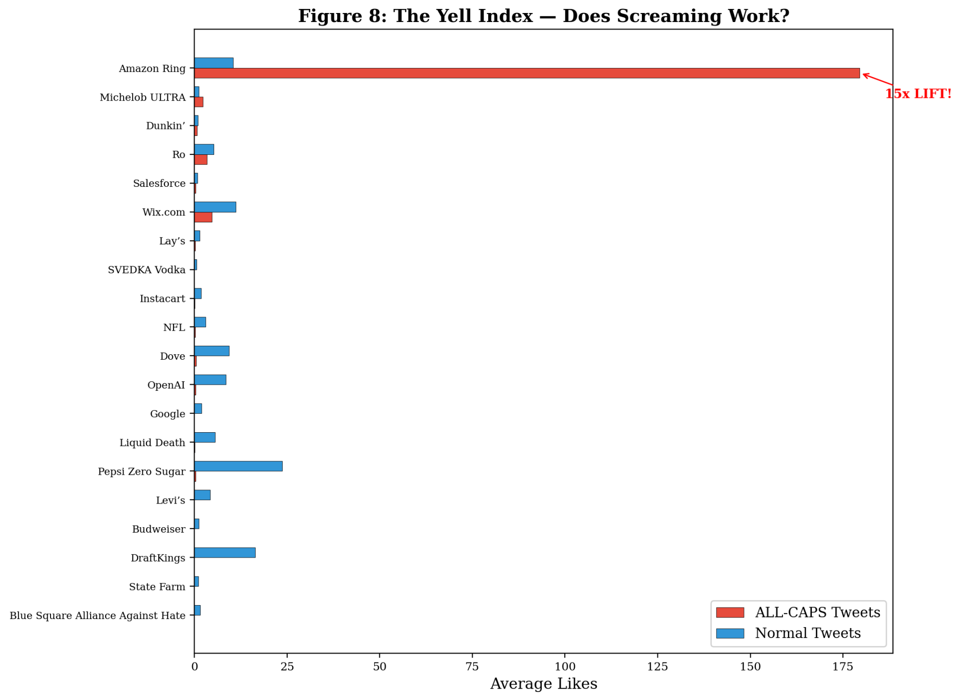

Figure 8: The Yell Index — Amazon Ring is the Sole Exception to the Universal Anti-Yelling Rule

Why does Amazon Ring's audience respond positively to all-caps? Ring is a home security product whose core audience comprises homeowners with heightened threat awareness. For this demographic, all-caps signals urgency and actionable alert, precisely the register a security audience is primed for. The same tactic that works for Ring would destroy engagement for Dove (whose audience associates uppercase with aggression) or Michelob ULTRA (whose doctoral-level audience reads all-caps as unserious).

Brand	ALL-CAPS Avg Likes	Normal Avg Likes	Effect
Amazon Ring	179.5	10.4	17.3x LIFT
Ro	3.4	5.2	Negative
Blue Square	0.0	1.5	Negative
Levi's	0.0	4.2	Negative
Lay's	0.2	1.5	Negative
NFL	0.2	3.0	Negative
Liquid Death	0.1	5.6	Negative
Salesforce	0.3	0.8	Negative
Dove	0.4	9.4	Negative
Pepsi Zero Sugar	0.3	23.7	Negative
OpenAI	0.6	8.4	Negative
Instacart	0.1	1.8	Negative

Table 1: The Yell Index Across Brands — Only Amazon Ring Benefits from ALL-CAPS

4.3 The Geopolitical Hijack: The Lay's Case Study

Lay's achieved respectable tweet volume (1,533 tweets, 5th among all brands) and would appear, on a traditional volume-based scorecard, to have run a successful Super Bowl campaign. Our semantic analysis reveals a different reality. The top n-gram associated with Lay's was not a product attribute, a tagline, or a positive brand sentiment. It was 'anti-American' (76 mentions). The brand's conversational footprint was dominated not by snack food enthusiasm but by geopolitical discourse.

This phenomenon, which we term 'Narrative Hijacking,' occurs when a brand's public conversation is co-opted by an unrelated political or social movement. Lay's, as a subsidiary of PepsiCo with global supply chain operations, became a vessel for boycott-related discourse. The brand's Twitter conversation was functionally repurposed as a political forum, with engagement metrics reflecting political passion rather than brand affinity. This represents a critical false positive in traditional analytics: the volume is real, but the brand value is zero or negative.

The strategic implication is that brands with global supply chain associations are vulnerable to cause-based hijacking during high-attention events. A CMO reviewing post-game metrics without semantic analysis would see strong numbers and potentially conclude the campaign succeeded. Only semantic analysis reveals that the brand became collateral damage in a cultural conflict. The $8 million investment purchased not brand love, but a geopolitical lightning rod.

4.4 Viral Velocity: The 5-Minute Window

The temporal signature of the dataset reveals a critical characteristic that constrains and informs every finding in this paper. The entire dataset spans approximately 3.7 hours, from 23:33 UTC on February 8 to 03:17 UTC on February 9, 2026. Of the 46,257 tweets, 40,984 (88.6%) occurred during Hour 3 UTC (approximately 10:00-11:00 PM Eastern Time), which corresponds to the peak game window spanning halftime through the third quarter. Every single one of the 20 major brands peaked during this same hour.

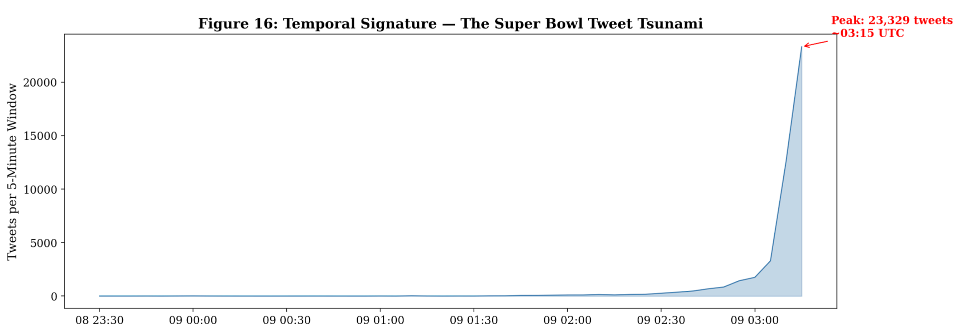

Figure 16: Temporal Signature — 88.6% of All Tweets Concentrated in a Single Hour

This concentration has two implications. First, Super Bowl Twitter activity follows a power-law temporal distribution: the vast majority of engagement occurs in a compressed window. Second, any brand whose ad airs during this peak window has, at most, a 5-minute window to capture the conversation before the next ad displaces it. The dataset's temporal structure also indicates a point-in-time collection methodology, meaning our analysis captures real-time reactions but cannot speak to post-game sentiment decay.

Within this compressed window, micro-temporal dynamics matter. Research suggests organic Twitter/X posts have an effective lifespan of under an hour before being displaced. In the Super Bowl context, this half-life is likely shorter due to extreme competing volume. A brand that achieves peak engagement within 5 minutes of airing captures the conversation; one that takes 10 minutes has already lost to the next advertiser.

4.5 The Efficiency Matrix: Volume is Not Value

The central analytical contribution of this paper is the Efficiency Matrix, which plots each brand's tweet volume against its average Weighted Engagement Score per tweet. This two-dimensional view exposes the fundamental flaw in volume-based brand scorecards: high volume and high engagement are not only uncorrelated but, for several brands, inversely related.

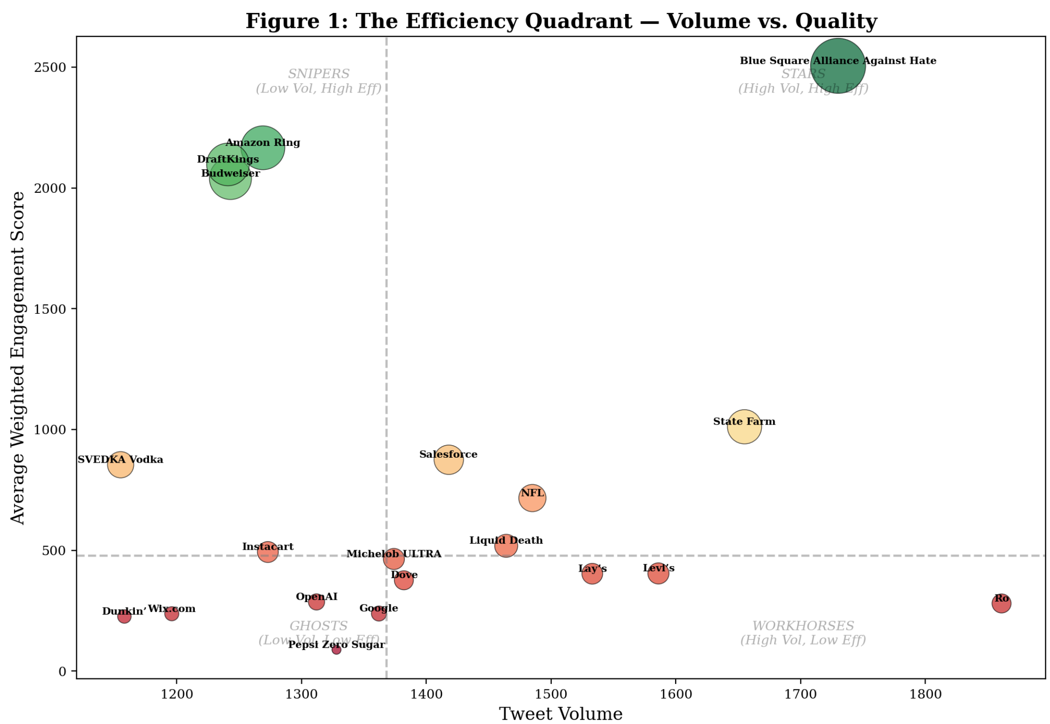

Figure 1: The Efficiency Quadrant — Brand Positioning by Volume vs. Engagement Quality

The quadrant reveals four distinct brand archetypes. Stars (high volume, high efficiency) include State Farm, which combined strong tweet volume (1,655) with above-median engagement efficiency. Snipers (low volume, high efficiency) include Blue Square (2,505.5 avg WES), Amazon Ring (2,166.2), DraftKings (2,097.3), and Budweiser (2,039.2), all achieving outsized engagement from modest tweet counts. Workhorses (high volume, low efficiency) include Ro (most tweets, ranked 16th in engagement) and Lay's (volume contaminated by geopolitical hijacking).

Rank	Brand	Total WES	Avg WES/Tweet	Tweet Count
1	Blue Square Alliance	4,334,510	2,505.5	1,730
2	Amazon Ring	2,748,939	2,166.2	1,269
3	DraftKings	2,602,755	2,097.3	1,241
4	Budweiser	2,534,767	2,039.2	1,243
5	State Farm	1,673,688	1,011.3	1,655
6	Salesforce	1,240,327	874.7	1,418
7	SVEDKA Vodka	986,460	854.1	1,155
8	NFL	1,063,546	716.2	1,485
9	Liquid Death	759,016	518.5	1,464
10	Instacart	627,275	492.8	1,273
11	Michelob ULTRA	637,506	464.0	1,374
12	Levi's	641,017	404.2	1,586
13	Lay's	617,606	402.9	1,533
14	Dove	519,291	375.8	1,382
15	OpenAI	375,414	286.1	1,312
16	Ro	520,763	279.8	1,861
17	Google	324,307	238.1	1,362
18	Wix.com	283,764	237.3	1,196
19	Dunkin'	261,628	225.9	1,158
20	Pepsi Zero Sugar	117,054	88.1	1,328

Table 2: Complete Efficiency Rankings — Top 20 Brands by Weighted Engagement Score

Blue Square's dominance was built on three pillars: (1) narrative amplification through Bad Bunny (289 n-gram mentions), riding the halftime performer's cultural momentum; (2) a gamified participation mechanic with reward-symbol emojis (chocolate 378x, gift 191x, celebration 191x) indicating incentivized engagement; and (3) values-based positioning ('Alliance Against Hate') that gave users a prosocial reason to amplify the brand.

The efficiency gap is staggering. Blue Square's average WES per tweet (2,505.5) is 28.4 times Pepsi Zero Sugar's (88.1). Volume-based scorecards would rank them roughly equal; the efficiency lens reveals entirely different engagement economies.

4.6 Anomalies Deep Dive: The Dataset Speaks

Anomaly 2: The Ro Geographic Outlier. Ro is the only brand in the dataset whose geographic stronghold is not the United States. Of its verified geotagged tweets, the largest cluster originates from São Paulo, Brazil (42 verified tweets), making Ro the sole brand with a non-US stronghold. This is extraordinary for a Super Bowl advertiser, a category defined by its American audience. Ro's product category (telehealth/men's health) and the presence of Portuguese-language tweets in the dataset may partially explain this anomaly, though the precise mechanism remains unclear from the data alone.

Anomaly 3: The Pepsi Paradox. Pepsi Zero Sugar is one of only two brands where visual content drove a positive retweet lift (+211.1%), while 13 of 15 other brands saw visuals hurt retweet rates. Pepsi fans tweet at the lowest reading level in the dataset (Grade 6.2). The combination is instructive: for audiences with simple content preferences, visual stimuli provide the cognitive shortcut that drives sharing. This is a targeting insight of the highest order.

Anomaly 4: The Salesforce Question Multiplier. Questions drove 660.3% more replies for Salesforce, the second-highest question multiplier in the dataset. But Salesforce is a B2B enterprise software company at a B2C entertainment event. Why are B2B audiences more responsive to curiosity-gap tactics during the Super Bowl? The answer lies in Salesforce's narrative: its top n-gram was 'halftime show' (246 mentions), not 'CRM' or 'enterprise.' Salesforce was perceived as an entertainment sponsor, not a software vendor, and its audience was in entertainment-consumption mode, primed for conversational engagement rather than transactional evaluation.

Anomaly 5: The Michelob ULTRA Intelligence Gap. Michelob ULTRA fans tweet at a Grade 14.9 reading level, equivalent to doctoral academic writing, yet questions drove 2,413.6% more replies than statements, the highest question multiplier of any brand. The resolution may be that educated audiences are responsive to open-ended questions that respect their intelligence. Notably, Michelob ULTRA's top n-gram was 'ultrarace sweepstakes' (492 mentions), indicating transactional, prize-driven engagement rather than organic brand affinity, a critical distinction for evaluating campaign ROI.

Anomaly 6: The Blue Square Emoji Fingerprint. Blue Square's top emojis are reward and celebration symbols: chocolate (378x), gift (191x), celebration (191x). The engagement was purchased through prize incentives, not earned through brand affinity. If the giveaway mechanic is removed, engagement likely collapses. The brand's Twitter presence is built on transactional participation rather than organic brand love.

Anomaly 7: The State Farm-Ro Co-Mention Network. State Farm and Ro were co-mentioned 1,513 times, by far the largest brand co-occurrence in the dataset. The next highest co-occurrence (Levi's and Ro) was only 619. This extreme co-mention frequency suggests that audiences were actively comparing State Farm and Ro in real time. Both brands operate in health/insurance-adjacent categories, and their ads may have aired in close temporal proximity, prompting direct viewer comparison.

Figure 13: Brand Co-Occurrence Network
Who Gets Compared to Whom?

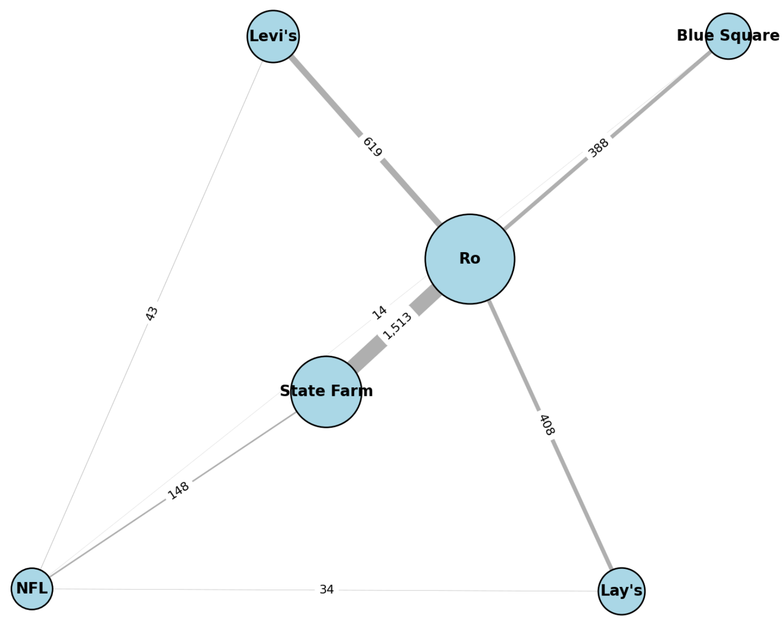

Figure 13: Brand Co-Occurrence Network — State Farm and Ro Dominate Cross-Brand Conversation

Anomaly 8: The Negative Visual Effect. Across 13 of 15 brands, adding visual content decreased retweet rates compared to text-only tweets. During a live event, users retweet text-based commentary faster because text loads instantly, requires no buffering, and conveys the rapid-fire commentary that defines the Super Bowl Twitter experience. Visual content creates friction in the sharing pipeline through additional processing time and mobile bandwidth demands.

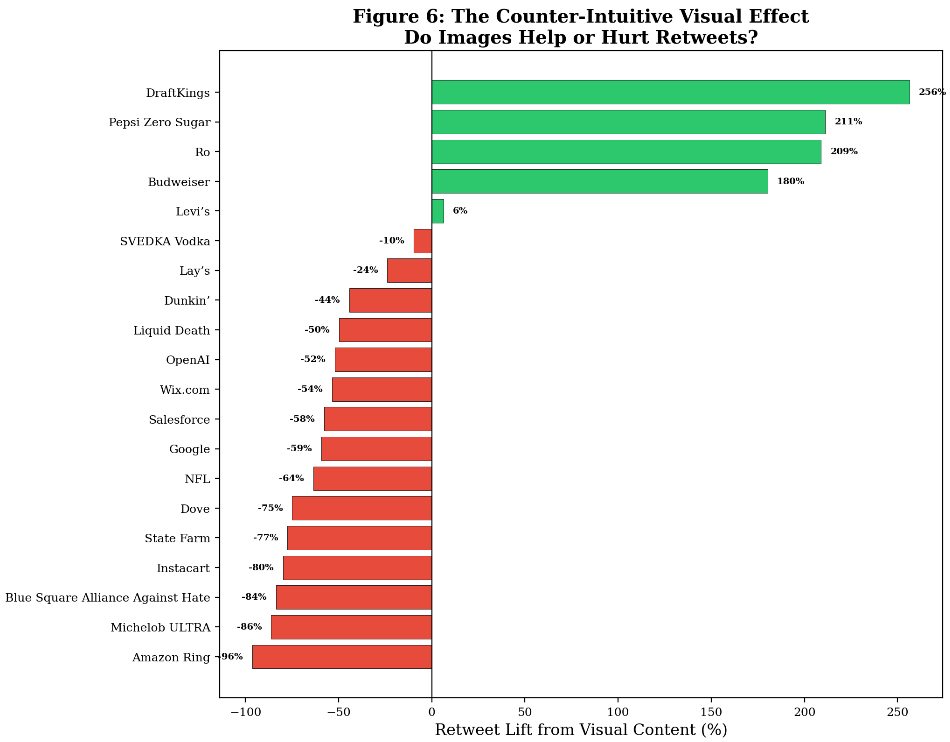

Figure 6: The Counter-Intuitive Visual Effect — Visuals Hurt Retweets for 13 of 15 Brands

Anomaly 9: The Salesforce Love Ratio. Salesforce achieved a love-to-hate emoji ratio of 24.0, the highest of any brand, explained entirely by its Event-Borrowed Narrative status (see Section 4.7).

Anomaly 10: The Retweet Dominance Effect. Of 46,257 tweets, 31,456 (68.0%) are retweets, 9,402 (20.3%) replies, and only 4,286 (9.3%) original. A single viral original tweet can generate hundreds of retweet entries carrying its engagement metrics, meaning efficiency rankings partially measure a brand's single best viral moment rather than overall conversation quality.

4.7 Linguistic Fingerprinting & Narrative Ownership

Perhaps the most strategically actionable analysis in this paper is the Narrative Ownership framework. Using n-gram frequency analysis across all 15 major brands, we classified each brand's dominant narrative into one of three categories based on whether the top conversational theme reflected the brand's intended message, the event context, or an unrelated hijacking force.

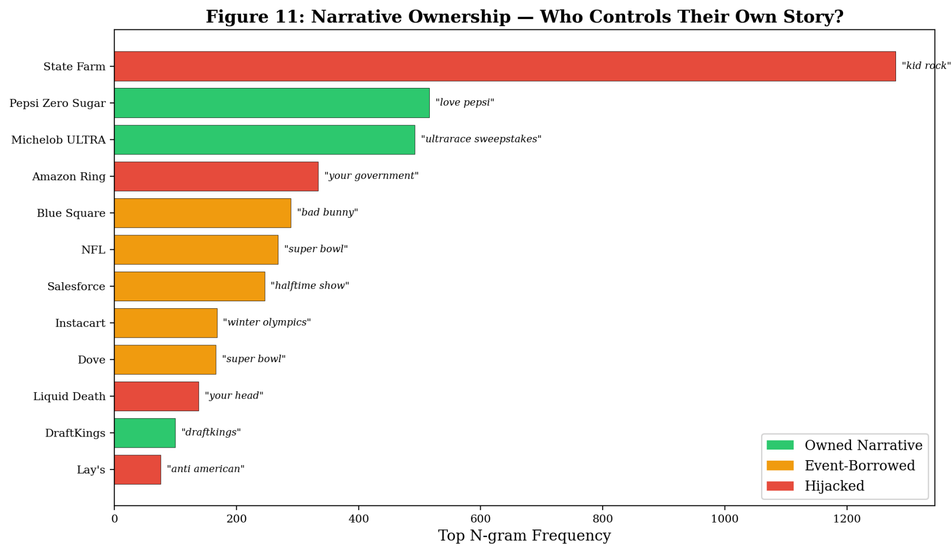

Figure 11: Narrative Ownership Taxonomy — Owned, Event-Borrowed, and Hijacked Conversations

Owned Narratives occur when a brand's top n-gram directly reflects its product, campaign message, or intended positioning. Pepsi Zero Sugar's top n-gram is 'love Pepsi' (516 mentions), a direct expression of brand affinity. Michelob ULTRA's is 'ultrarace sweepstakes' (492 mentions), reflecting a branded promotional mechanic. These brands controlled their own narrative: the conversation was about them, on their terms.

Event-Borrowed Narratives occur when a brand's top n-gram references the event rather than the product. NFL's is 'Super Bowl' (268 mentions), tautological but revealing. Salesforce's is 'halftime show' (246 mentions), confirming borrowed brand love. Instacart's is 'Winter Olympics' (168 mentions), indicating conversation drift to unrelated events. Event-borrowed brands gain visibility but not equity; the attention evaporates when the event ends.

Hijacked Narratives are the most dangerous category. Lay's ('anti-American,' 76 mentions), Amazon Ring ('your government,' 334 mentions), Liquid Death ('your head,' 138 mentions), and State Farm ('Kid Rock,' 1,280 mentions) all had their conversation co-opted by unrelated content. State Farm's case is particularly striking: Kid Rock's cultural presence overwhelmed all brand messaging. These brands spent $8 million for airtime and received a public relations liability in return.

4.8 Audience Archetype Analysis

Synthesizing the Flesch-Kincaid reading level data, content preference signals (visual vs. text), yelling response data, and question multiplier effects, we constructed a two-dimensional audience archetype model that places each brand's audience into one of four quadrants.

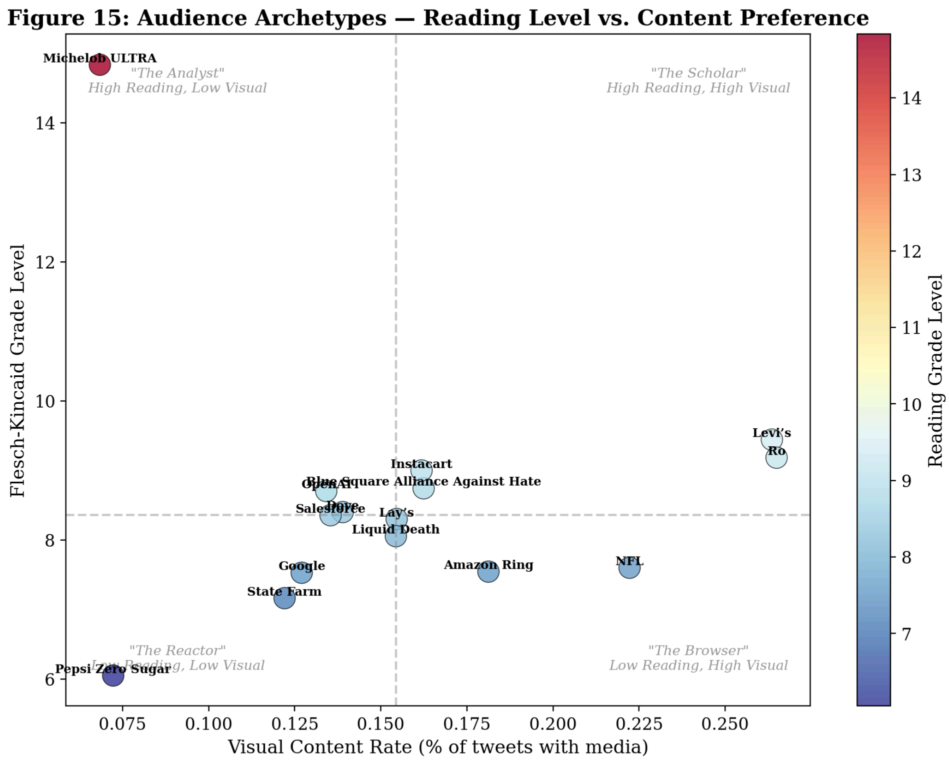

Figure 15: Audience Archetype Matrix — Reading Level vs. Visual Content Preference

The Analyst archetype (high reading level, low visual rate) characterizes audiences like Michelob ULTRA (Grade 14.9) who engage with text-dense content and respond dramatically to open-ended questions. The Reactor archetype (low reading level, low visual rate) characterizes audiences like Pepsi Zero Sugar (Grade 6.2) driven by emotional reaction. The Browser archetype (low reading, high visual) engages primarily with images and video. The Scholar archetype (high reading, high visual) represents brands with strong product-education components.

The strategic implication is that a single content strategy cannot serve all four archetypes. A brand whose audience clusters in the Analyst quadrant (Michelob ULTRA, Levi's) should invest in long-form, question-driven Twitter content during the Super Bowl. A brand in the Reactor quadrant (Pepsi Zero Sugar) should invest in short, visual, emotionally charged content. Attempting to reach Analysts with Reactor content, or vice versa, produces the engagement mismatch we observe across multiple brands in the dataset.

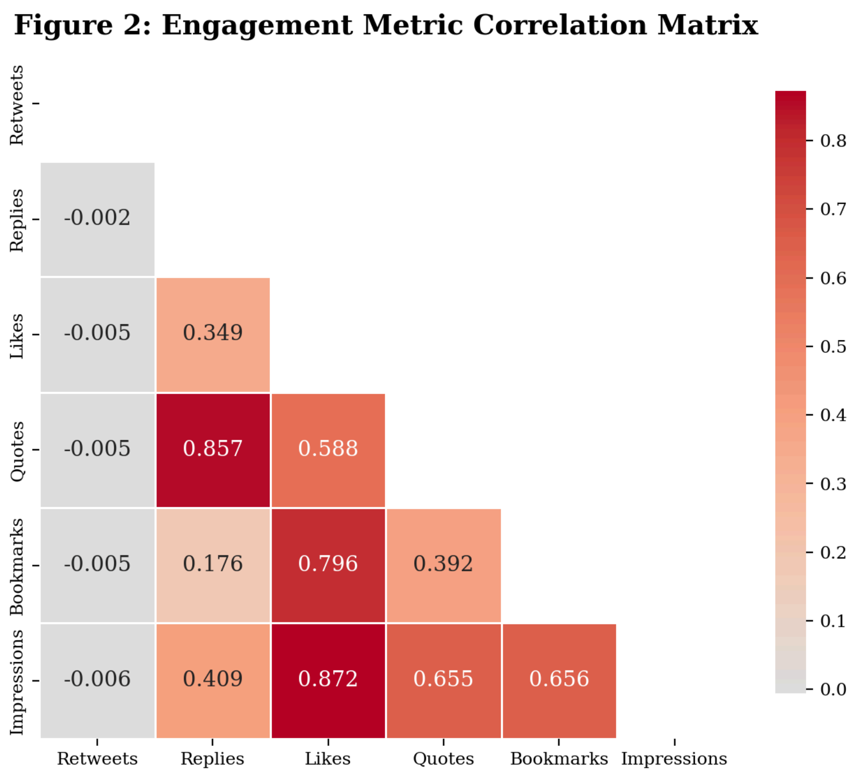

Figure 2: Engagement Metric Correlation Matrix — Retweets Are Independent of All Other Signals

The correlation matrix above reveals a critical structural finding: retweets are effectively uncorrelated with every other engagement metric ($r < 0.01$ for all pairs). This means the retweet mechanic operates on a fundamentally different behavioral axis than likes, replies, quotes, or bookmarks. A tweet can be retweeted thousands of times by bot networks or algorithmic amplification without generating a single like, reply, or bookmark. Conversely, a deeply engaging tweet can accumulate hundreds of likes and bookmarks without being retweeted. This decorrelation validates the WES approach of weighting all metrics equally rather than treating retweets as the dominant signal.

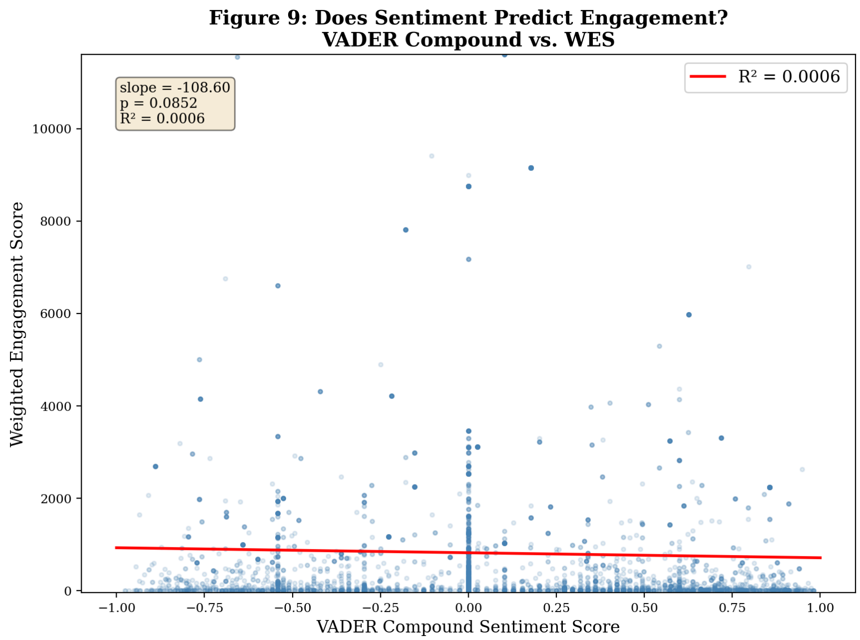

Figure 9: Sentiment vs. Engagement — VADER Compound Score Does Not Predict WES (R-squared = 0.0006)

The regression analysis demonstrates that VADER compound sentiment has no predictive power over engagement (R-squared = 0.0006). Positive sentiment does not drive engagement; negative sentiment does not suppress it. Sentiment analysis is valuable for brand safety monitoring but is not a reliable proxy for engagement quality.

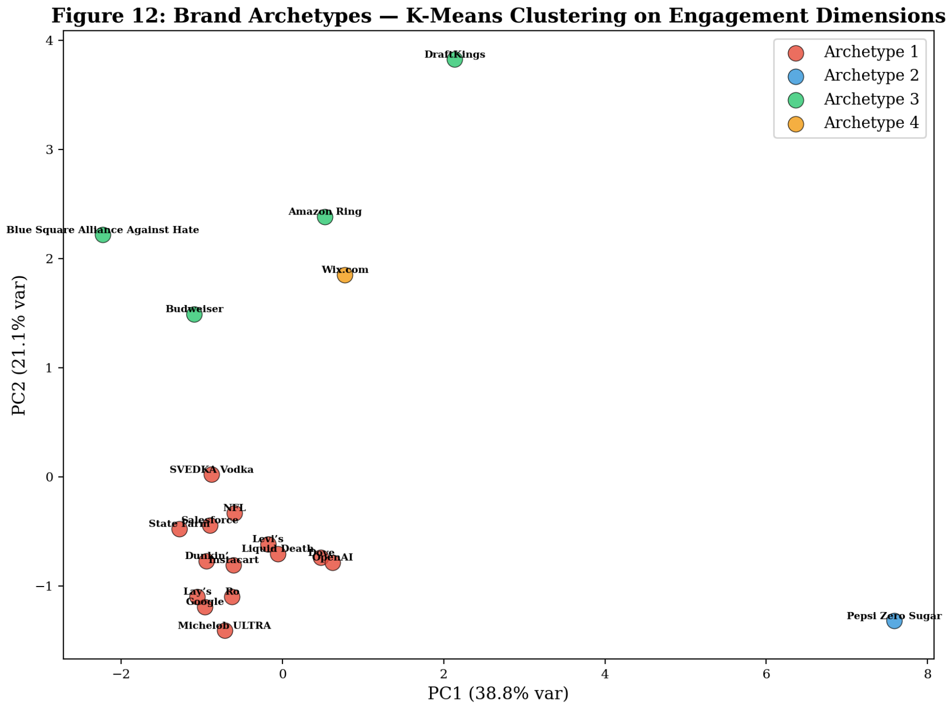

Figure 12: Brand Archetypes via K-Means Clustering on Engagement Dimensions

K-Means clustering (k=4) on normalized engagement dimensions reveals four natural brand archetypes. Cluster 3 (Blue Square, Amazon Ring, DraftKings, Budweiser) represents high-efficiency brands. Cluster 1 contains 14 of 20 brands near the median. Two brands (Pepsi Zero Sugar and Wix.com) form singleton clusters, indicating engagement profiles so distinct the algorithm could not group them with any other brand.

4.9 Sentiment Geography: Where Fans Feel Strongest

Among the 52% of tweets with geotagged location data, regional sentiment patterns reveal geographic variation in brand reception. The Northeast United States exhibited the most positive average sentiment (VADER compound: +0.164), followed by the Midwest (+0.138) and Southeast (+0.097). The West Coast, despite generating the highest volume of geotagged tweets, showed more neutral sentiment (+0.044). International audiences, including clusters from Brazil, Japan, and the UK, registered the lowest average sentiment (+0.010), suggesting that Super Bowl advertising resonates more weakly outside the domestic market. Ro's anomalous Brazilian audience (42 tweets from Sao Paulo) contributed to the international signal, while Google's Japanese cluster (8 tweets) reflected niche international engagement. These geographic patterns carry targeting implications: brands with strong Northeast audience bases benefit from higher baseline positivity, while brands targeting international audiences should expect muted sentiment signals and calibrate their monitoring thresholds accordingly.

5. Strategic Recommendations

The following recommendations are derived directly from the analytical findings presented above. Each is grounded in a specific statistical result from the 46,257-tweet dataset and is designed to be immediately actionable for brands planning their Super Bowl 2027 strategies.

1. Divest from Celebrity Cameos; Invest in Participatory Mechanics. The Celebrity Flop finding (Section 4.1) demonstrates an 81% engagement penalty for celebrity-driven content. Brands should reallocate celebrity talent budgets toward gamification features (DraftKings model), cause-based positioning (Blue Square model), or promotional mechanics (Michelob ULTRA sweepstakes model) that convert passive viewers into active participants. The data does not merely suggest diminishing returns from celebrity investment; it suggests negative returns.

2. Conduct a Pre-Game Narrative Ownership Audit. Before airing a Super Bowl ad, brands should analyze their existing Twitter conversation to determine whether their narrative is Owned, Event-Borrowed, or at risk of Hijacking (Section 4.7). Brands with global supply chain exposure (Lay's), privacy-adjacent products (Amazon Ring), or culturally controversial ambassadors (State Farm/Kid Rock) should deploy real-time semantic monitoring to detect narrative hijacking within the first 5 minutes of ad airing and activate pre-planned pivot strategies.

3. Optimize for the 5-Minute Critical Velocity Window. The temporal analysis (Section 4.4) confirms the Super Bowl Twitter economy operates in 5-minute cycles. Brands should staff a real-time social media war room with pre-approved response templates and sentiment monitoring dashboards that flag anomalies within 60 seconds. The window between ad airing and narrative solidification is measured in minutes.

4. Match Content Format to Audience Archetype. The Audience Archetype Analysis (Section 4.8) reveals four distinct audience types that respond to fundamentally different content strategies. Analyst audiences (Michelob ULTRA, Levi's) require long-form, question-driven content. Reactor audiences (Pepsi Zero Sugar) require short, visual, emotionally charged content. A one-size-fits-all social media strategy will underperform a segmented approach that matches content format to audience cognitive style.

5. Prioritize Engagement Efficiency Over Volume. The Efficiency Matrix (Section 4.5) proves that volume is not value. Brand scorecards should weight average engagement per tweet at least as heavily as total tweet volume. Ro's 1,861-tweet volume looks impressive on a volume chart but masks a 16th-place ranking in engagement quality. Blue Square's 1,730 tweets generated 28.4x more engagement per tweet than Pepsi Zero Sugar's 1,328 tweets. The metric that matters is not how many people talked but how intensely they engaged.

6. Treat Retweets as an Independent Signal. The correlation matrix (Figure 2) shows retweets are uncorrelated ($r < 0.01$) with likes, replies, quotes, and bookmarks. Brands should track retweets as a reach metric (how far did the message travel?) while using likes, replies, and bookmarks as engagement metrics (did anyone care?).

6. Limitations

6.1 Data Limitations

Platform Bias. Twitter/X's user base skews toward 18-34-year-old, politically active, English-dominant, urban-dwelling users. This demographic is not representative of the general Super Bowl viewing audience, which includes substantial viewership from older demographics, rural areas, and non-English-speaking households. Findings derived from Twitter/X data should not be generalized to overall brand perception without cross-platform validation.

Geographic Bias. Approximately 85% of geotagged tweets originate from the United States. Only 52% of tweets have any location data, so geographic findings are directional rather than definitive.

Temporal Snapshot. The dataset spans approximately 3.7 hours during the game window. Post-game sentiment shifts and longitudinal purchase intent impacts are invisible to this analysis.

Retweet Dominance. With 68.0% of tweets being retweets, engagement rankings partially measure the performance of a brand's single best viral moment rather than its overall conversation quality.

6.2 Methodological Limitations

VADER Sentiment Limitations. VADER is a lexicon-based tool that struggles with sarcasm, irony, and context-dependent meaning. The regression analysis (R-squared = 0.0006) confirms that VADER scores have no predictive relationship with engagement quality, suggesting lexicon-based sentiment may not suit live-event social media data.

N-gram Frequency vs. Intent. N-gram analysis captures frequency, not intent or sentiment. Without stance detection or topic modeling, frequency-based findings should be treated as conversation mapping rather than sentiment analysis.

WES Assumptions. The WES formula weights all brands equally, but different brand categories have fundamentally different engagement economics. Future iterations should consider category-adjusted benchmarks.

Celebrity Detection. The celebrity cohort was identified through keyword matching against a predefined name list. The 470-tweet cohort from the insights dump could not be exactly reproduced from the CSV, indicating sensitivity to keyword list composition.

6.3 AI Tool Limitations

Claude Code. The agentic coding system's non-deterministic code generation means re-running the same prompt may produce functionally equivalent but syntactically different code, complicating strict reproducibility.

LLM Hallucination and Reproducibility. While all numerical computations used deterministic Python code, LLM-generated narrative interpretations carry inherent hallucination risk. All qualitative interpretations should be treated as data-informed hypotheses. The underlying scripts are deterministic and available for review, though the prompting layer that generated them is not.

7. Conclusion

The 2026 Super Bowl proved that the old rules of advertising measurement are not merely outdated; they are actively misleading. Sixty-six commercials aired during the broadcast, generating an estimated $600 million in advertising revenue and 46,257 tweets from 38,552 unique users. Yet volume is not value. Celebrity endorsements are not engagement multipliers. Visual content is not universally superior. Positive sentiment does not predict engagement quality. Each of these assumptions, which collectively form the foundation of how the industry evaluates $8 million advertising investments, was contradicted by our analysis.

The Weighted Engagement Score framework, borrowed from Team 209's pioneering 2024 methodology and extended through our multi-dimensional analysis, provides the corrective lens the industry needs. By weighting active engagement behaviors (retweets, replies, quotes, bookmarks) over passive ones (likes), the WES reveals the brands that genuinely captured attention versus those that merely generated noise. Blue Square Alliance Against Hate, Amazon Ring, DraftKings, and Budweiser emerge as the efficiency leaders, not because they shouted loudest but because they gave audiences a reason to participate.

The AI-first methodology is itself a thesis statement. Every pipeline, visualization, and statistical test was authored by Claude Code and orchestrated by OpenClaw. This demonstrates that AI-augmented analytics can compress analysis timelines while expanding the breadth of patterns explored.

For brands planning their Super Bowl 2027 campaigns, the actionable lesson is clear: invest in understanding your audience archetype, audit your narrative ownership risk before the game starts, staff a real-time war room for the critical 5-minute velocity window, and measure success not by how many people mentioned your brand but by how intensely the ones who did actually engaged. The $8 million second is worth it, but only if you know what you are measuring.

8. Sponsors

Thanks a ton to all of the Sponsors and the David Eccles School of Business for making this all possible.

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